Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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PROFESSIONAL PROFILE

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

AREAS OF SPECIALISATION

R&D in avionics systems (12 years)
Twelve years of mission-computer and sensor integration, interface testing (MIL-STD-1553, ARINC 429) and environmental qualification for airborne platforms at Halcyon Avionics Design Bureau.

Edge AI for aerial imagery (5 years)
Research on super-resolution and model compression (pruning, knowledge distillation) for real-time deployment on NVIDIA Jetson devices; publications and a deployed telemetry prototype.

Upper-air instrumentation (10 years)
Planning, testing and maintenance of radiosonde ground systems and upper-air instruments; division workshop management; technical evaluation of procurements.

IoT monitoring systems (4 years)
Design of MQTT, Node-RED, InfluxDB and Grafana pipelines for monitoring ground-station equipment.

TECHNICAL DOMAINS

Avionics (MIL-STD-1553 / ARINC 429, Avionics integration) · Scientific instrumentation (Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems) · Embedded systems (Embedded C, ESP32 / STM32 microcontrollers) · Computer vision (Image super-resolution, Object detection) · Drones & UAV systems (UAV payload integration) · Edge AI (NVIDIA Jetson deployment) · AI / machine learning (Deep learning, PyTorch) · Procurement & administration (Technical procurement (GeM, GFR)) · IoT (MQTT and Node-RED, InfluxDB and Grafana) · Model optimisation (TensorRT and ONNX, Pruning and knowledge distillation) · Software & tools (Python, Docker and Linux) · Technical leadership (Team leadership) · GPU computing (CUDA / GPU computing)

JUSTIFICATION AGAINST THE REQUIREMENTS OF THE POST

Essential: Master's degree in Electronics / Electronics and Communication / Avionics Engineering from a recognised university.
Evidence: M.Tech., Avionics — Coastal Institute of Aeronautical Engineering (2010)

Essential: At least 10 years of experience in design, integration or testing of avionics, embedded or sensor systems.
Evidence: At least 23 years (since Aug 2002) in avionics, test & evaluation: Scientist E, Kestrel Atmospheric Instrumentation Laboratory; Senior Design Engineer, Halcyon Avionics Design Bureau; Graduate Engineer, Meridian Electronics Limited

Desirable: Ph.D. in Electronics, Electrical or Computer Engineering.
Evidence: Ph.D., Electrical and Electronics Engineering — Northfield Institute of Science and Technology (ongoing)

Desirable: Experience in machine learning, computer vision or edge AI deployment for airborne or UAV platforms.
Evidence: Skill: Image super-resolution; Skill: Object detection; Project: Edge super-resolution pipeline for UAV imagery

Fictional sample profile — not a real person

Desirable: Publications in peer-reviewed journals.
Evidence: 6 peer-reviewed publication(s): Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs; An IoT telemetry framework for upper-air ground stations; A low-cost GNSS receiver chain for radiosonde tracking

Desirable: Experience of leading technical teams and project coordination.
Evidence: Project: Radiosonde ground-station acceptance test protocol; Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results

Desirable: Design and evaluation of airborne sensor and avionics subsystems
Evidence: Developed LabVIEW test sequences for power-supply and interface boards of avionics units; Mentored graduate engineer trainees on avionics bus protocols and test documentation; Project: IoT monitoring of laboratory infrastructure

Desirable: development of embedded processing for UAV payloads
Evidence: Project: Edge super-resolution pipeline for UAV imagery; Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment; Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT

Desirable: preparation of technical specifications and test plans
Evidence: Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules; Scientist E, Kestrel Atmospheric Instrumentation Laboratory

Desirable: guidance of junior scientists.
Evidence: Publication: A low-cost GNSS receiver chain for radiosonde tracking; Scientist E, Kestrel Atmospheric Instrumentation Laboratory

POSITIONS HELD

Scientist E Jul 2016 – Present

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

Tools: Python, PyTorch, MQTT, Node-RED, InfluxDB, Grafana, Docker

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Supervise technicians and plan preventive maintenance for workshop test equipment.
- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

Senior Design Engineer Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

Tools: MIL-STD-1553, ARINC 429, LabVIEW, C, MATLAB

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

Fictional sample profile — not a real person

- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

Graduate Engineer Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

Tools: LabVIEW, C

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

MAJOR R&D PROJECTS AND TECHNICAL CONTRIBUTIONS

Edge super-resolution pipeline for UAV imagery *(Ongoing)* Sep 2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Radiosonde ground-station acceptance test protocol Jan 2018 – Mar 2019

Coordinator

Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure Feb 2020 – Jun 2022

Project lead

Instrument rooms lacked continuous environmental and power-quality monitoring.

Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

LEADERSHIP, COMMITTEES AND RESPONSIBILITIES

Supervisor, M.Tech. interns in applied machine learning 2022 – Present

Kestrel Atmospheric Instrumentation Laboratory

Supervise summer and dissertation interns on edge-AI projects.

Chair, Technical Evaluation Committee for radiosonde procurement Apr 2023 – Feb 2024

Kestrel Atmospheric Instrumentation Laboratory

Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, “Knowledge-distilled lightweight networks for drone image super-resolution,” accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, “Benchmarking embedded GPUs for real-time image enhancement,” Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

AWARDS & HONOURS

Best Paper Award 2023
Example Conference on Instrumentation (ECI 2023)

RESEARCH GRANTS & FUNDING

Seed grant: edge-AI enhancement of aerial imagery *(Completed)* Apr 2023 – Mar 2025
Principal investigator, Example Science Foundation (fictional)
Two-year seed project on lightweight super-resolution networks for embedded GPUs.
Reference: ESF/SEED/2023/017

TALKS & PRESENTATIONS

Invited talk: Edge AI for atmospheric observation 2024
Example Symposium on Applied AI

TRAINING PROGRAMMES ATTENDED

Public Procurement and GFR 2017 Nov 2019
Example Institute of Public Administration (sponsored by Kestrel Atmospheric Instrumentation Laboratory)
Duration: 5 days
Skills: Public procurement, GFR 2017, GeM

Mid-Career Training Programme for Scientists Feb 2020
Example Institute of Scientific Administration (sponsored by Ministry of Example Sciences)
Duration: 10 days

CERTIFICATIONS & LICENCES

Deep Learning for Edge Devices Mar 2022
Example Academy of Applied Computing
Used the model-compression techniques to deploy the drone super-resolution model on Jetson modules.
Duration: 40 hours
Credential: EAAC-DL-0042
Valid until: Mar 2029
Mode: Online
Skills: Deep learning, Edge AI, TensorRT

Academic English proficiency test *(Expired)* May 2016
Example Language Testing Board
Result: Overall band 8.0

Expired: May 2018

PROFESSIONAL COURSES

Introduction to Machine Learning (12-week course) Oct 2018
Example Online University Consortium
Duration: 48 hours
Result: Elite (82%)
Mode: Online
Skills: Machine learning, Python

WORKSHOPS, SEMINARS & CONFERENCES ATTENDED

Workshop on Efficient Deep Learning for Embedded Vision Dec 2023
Example Conference on Instrumentation (ECI 2023)
Duration: 2 days

National Seminar on Upper-Air Observing Systems Aug 2021
Example Society of Atmospheric Instrumentation
Duration: 1 day

EDUCATION

Ph.D., Electrical and Electronics Engineering 2021 – ongoing
Northfield Institute of Science and Technology
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics 2010
Coastal Institute of Aeronautical Engineering, Chennai
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering 2004
Northfield Institute of Science and Technology, Northfield University, Pune
Grade: First Class with Distinction

PROFESSIONAL MEMBERSHIPS

Member, Institution of Example Engineers 2010 – Present
Institution of Example Engineers

PROFESSIONAL SERVICE

Reviewer, Example Journal of Atmospheric Instrumentation 2022 – Present
Example Journal of Atmospheric Instrumentation

Co-supervisor, two M.Tech. dissertations in embedded vision 2024 – 2025
Example Institute of Technology (fictional)

LANGUAGES

English, Hindi

REFEREES

Dr. R. Iyer (fictional)
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